Exploring quantum mechanics with Qiskit



One Challange to rule them all

Qiskit Fall Fest 2026
IISER Kolkata

One Quantum Challenge To Rule Them All

The challenge

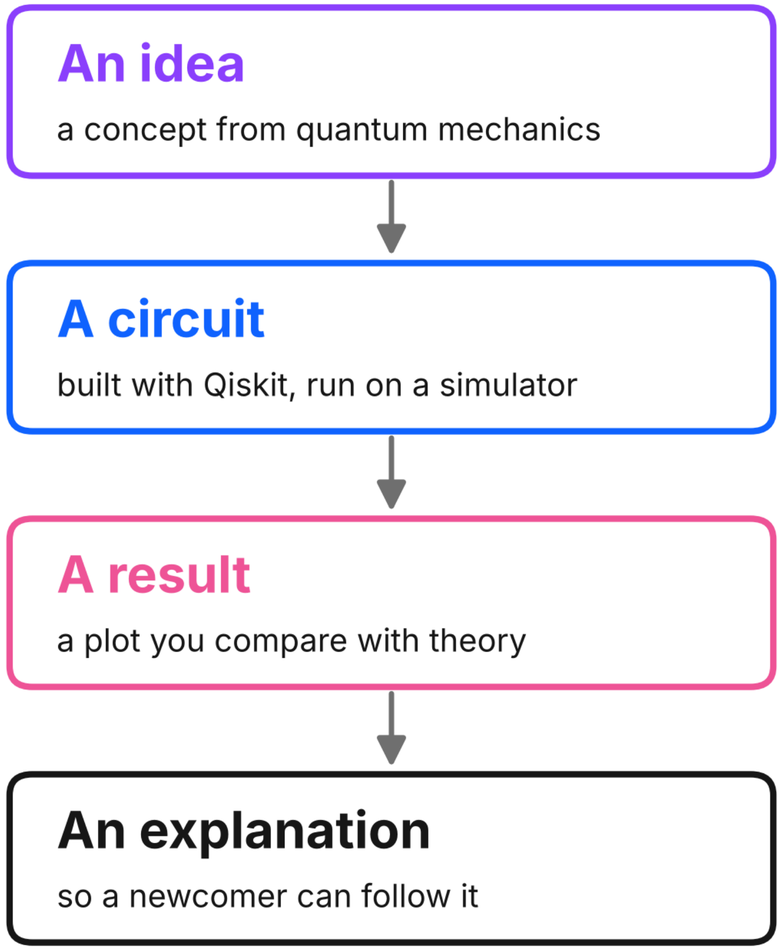
Pick one idea in quantum mechanics.

Design a demonstration that uses Qiskit to illustrate it.

Individuals or groups.

Any Python module is allowed. Qiskit must appear in your presentation.

The top 3 projects win prizes.



Start familiar, then go further

Familiar starting points	Where we want you to go
Entanglement	A novel or less commonly shown idea.
Teleportation	Or a fresh perspective on a known concept.
Bell's theorem	



Find your level

Starter

Sequential Stern-Gerlach. Two-path interference. Rabi oscillations.

Stretch

Bell and CHSH under noise. Quantum Zeno effect. Decoherence.

Bold

Adiabatic theorem. Berry phase. Tunnelling. Contextuality.

Your own

Anything not on this list. Ask us if you are unsure.

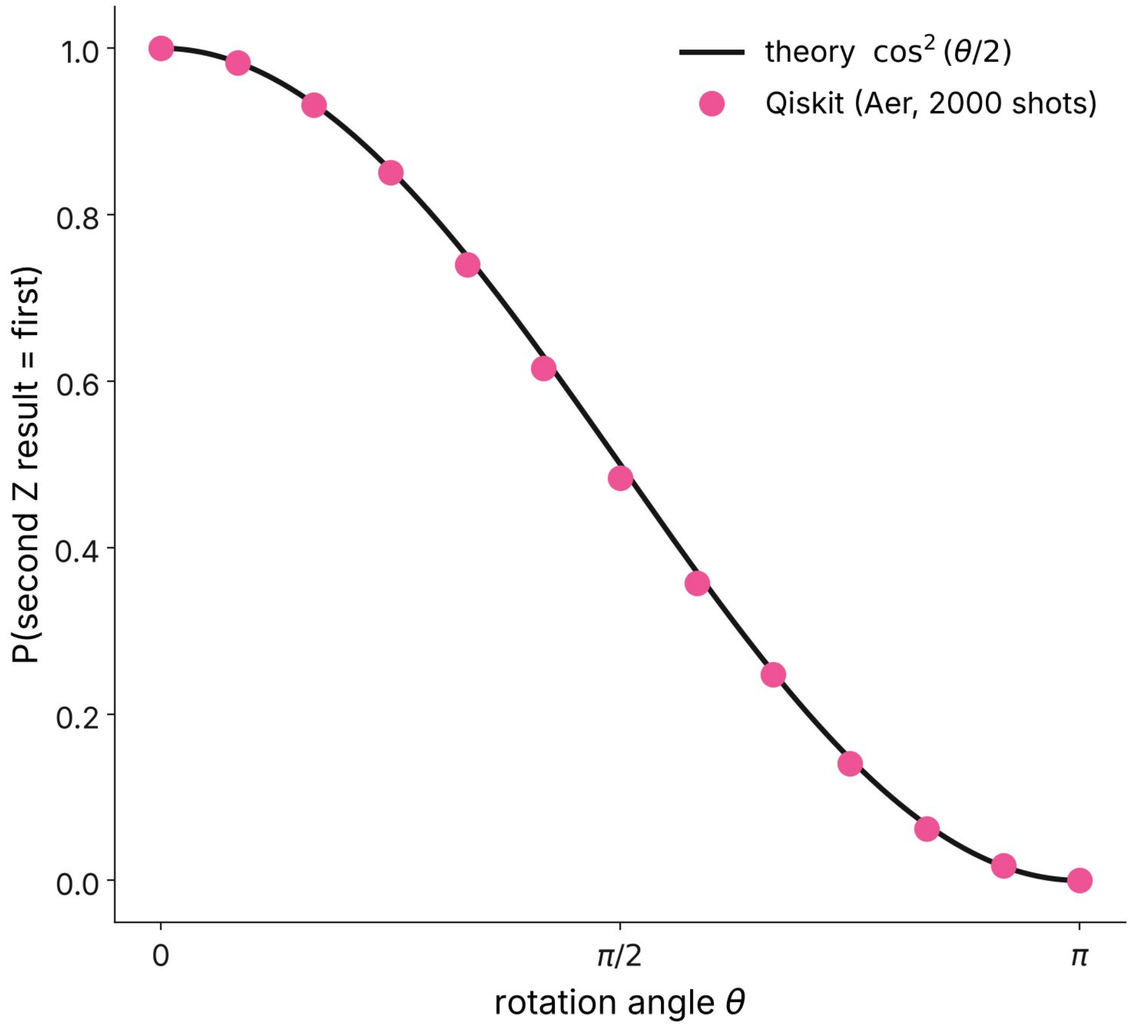
Starter example: sequential Stern-Gerlach

Measure Z, rotate the axis by θ, measure Z again.

Does the second result match the first?

Quantum prediction: $\cos^2(\theta/2)$.

The Qiskit points sit on the theory curve, within shot noise.



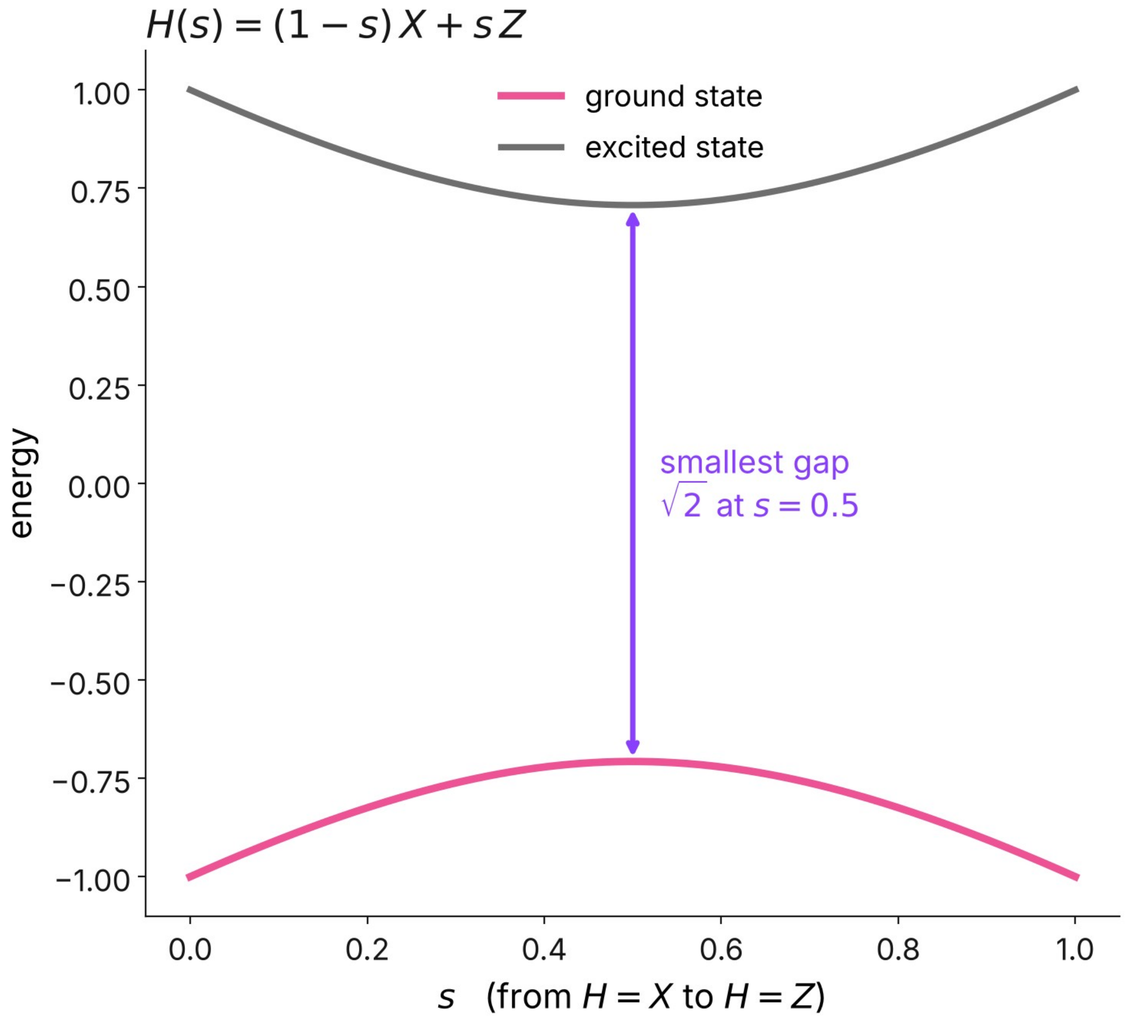
Example entry: the adiabatic theorem

Question: if a Hamiltonian changes slowly enough, does a qubit stay in its ground state?

H(s) = (1 − s) X + s Z, from s = 0 to s = 1 over a total time T.

Start in the ground state of X.

The energy gap is smallest at s = 0.5. That is where going too fast hurts.



Example entry: the circuit

Split the time T into N small steps.

Each step is one X rotation and one Z rotation.

This is a Trotterised version of the evolution.

At the end, measure: is the qubit in the ground state of Z?

1. Prepare



start in the ground state of X

2. Repeat N times (step k)



$\Delta t = T/N,\ s_k = (k - \frac{1}{2})/N,\ k = 1 \ldots N$

3. Measure



is it in the ground state of Z?

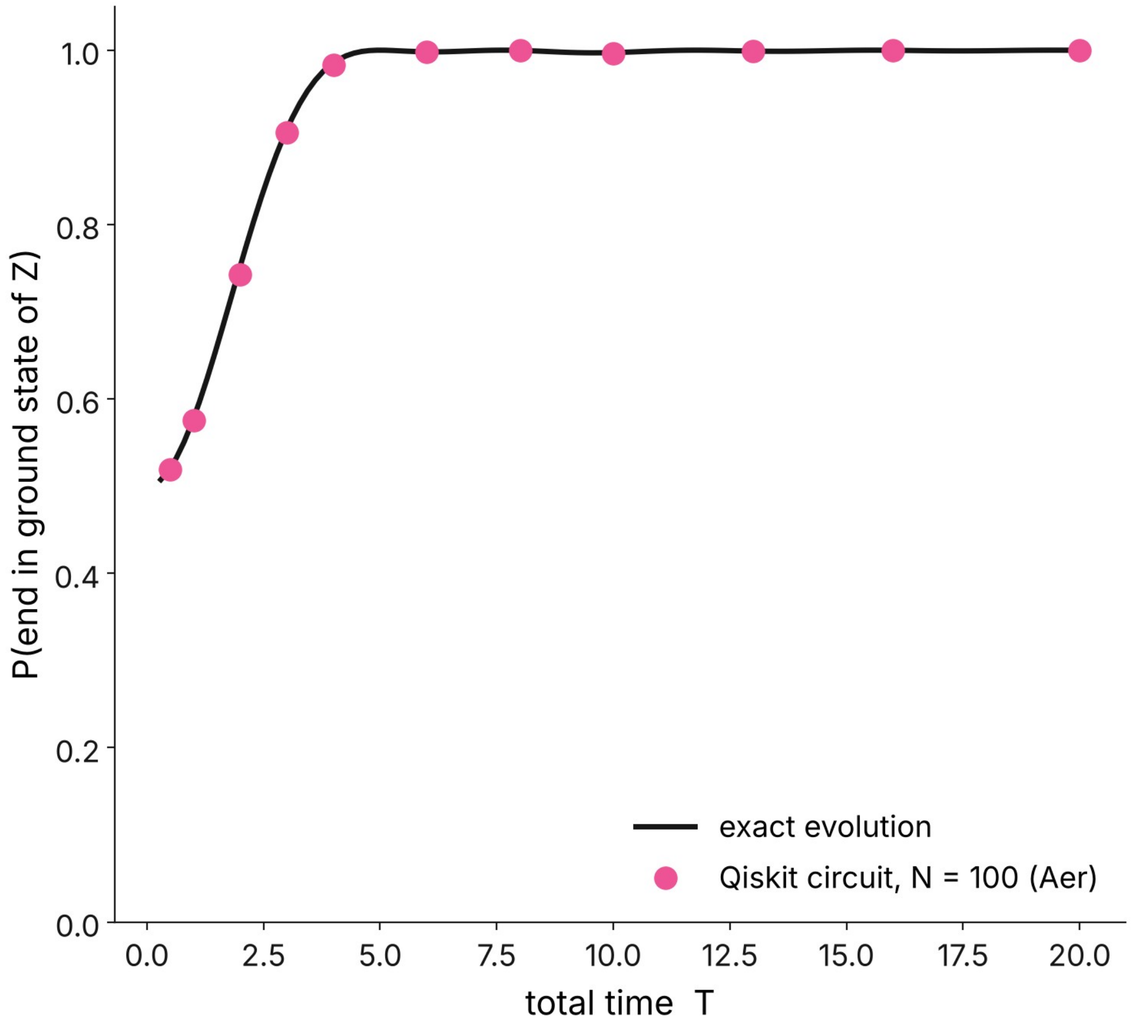
Example entry: the result

Run on Aer for many values of T.

Plot the probability of ending in the ground state of Z.

Short T: the state barely moves, so we get 50 percent.

Long T: the qubit follows the ground state, and we get close to 100 percent.



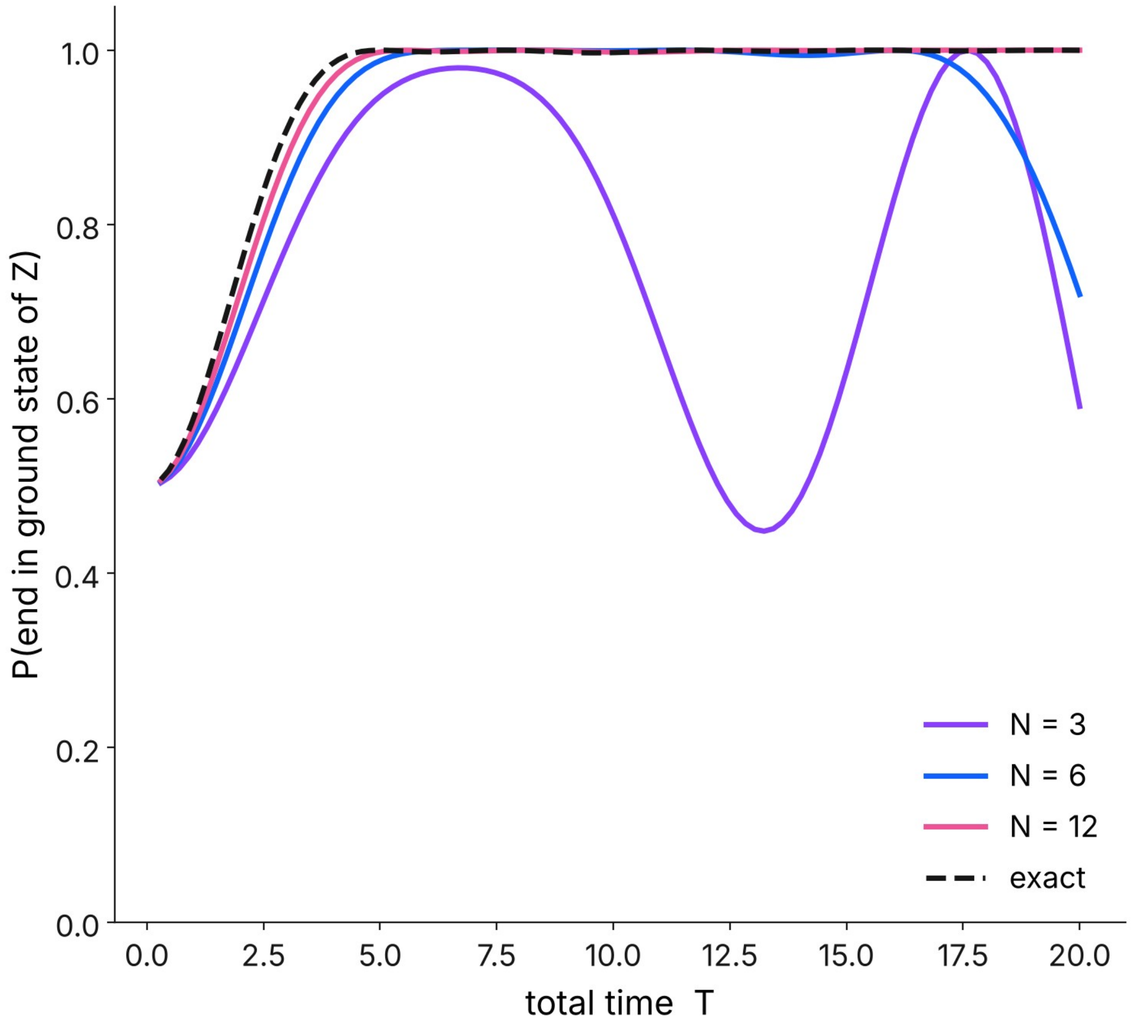
Example entry: check and limits

Checked against an exact calculation.

The needed T is set by the energy gap.

Too few steps N gives errors, so N must grow with T.

Finite shots add noise to every point.



Any Python module is allowed, from NumPy to Streamlit.

Any

The one requirement: Qiskit appears in your presentation.

Qiskit

Judged on three things

35	Creativity: a fresh idea, or a fresh angle on a known one.
30	Clarity: a newcomer can follow it.
35	Technical soundness: the physics and the circuits are right.

Points out of 100. The top 3 projects win prizes.

What to hand in, and when

- A notebook.

- A short presentation or live demo.

- One paragraph on the idea you chose.

Submit by the end of October. [Via Given Google Form]

Exact date to submit: to be announced.

Pick an idea

choose a concept and one question

Build with Qiskit

circuit, simulation, plot

Check against theory

does the result match?

Submit

end of October



Qiskit Fall Fest
Swag

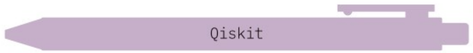
Notebook



Card Case



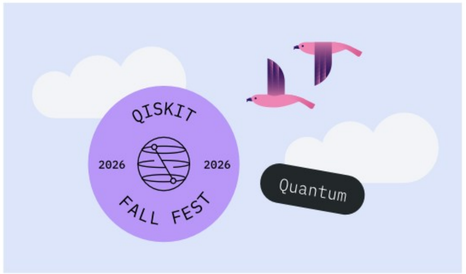
Pen



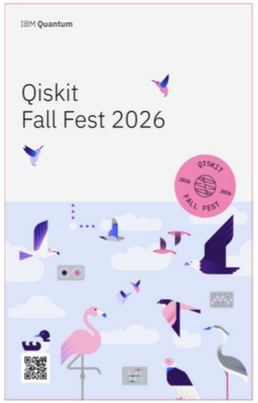
Cap



Magent



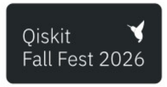
Poster



Lunch Cooler Bag



Stickers





Now, pick your idea

